Cardio Digital Twin — Video Script

Priyanka Bharwani · solo entry · final version · 17 October 2026

How to use this script

Eleven clips, about 18½ minutes in total, inside the 15–20 minute limit. Record each clip as its own file and join them at the end. Slide numbers match the updated 16-slide deck.

#	Clip	On screen	Length
1	Intro	Your face, full screen	1:00
2	Problem and use case	Slides 1–3	1:45
3	Data, data flow and architecture	Slides 4–6	2:00
4	The app at a glance	Slide 7	0:30
5	ECG model and results	Slides 8–9	2:15
6	Wearable simulation and monitor	Slides 10–11	2:00
7	The twin learns from day one	Slide 12	1:15
8	The twin: combining the streams	Slide 13	1:00
9	Live demo of DilSe	The app in your browser	4:15
10	Integrity, limits, next steps	Slides 14–16	1:45
11	Close	Your face, full screen	0:45

- **Plain paragraphs** are what you say. **Lines starting with "Screen:"** are what to show; don't read them out.
- Speak a little slower than feels natural (about 140 words a minute). It sounds right on playback.
- Made a mistake? Pause for 2 seconds and repeat the sentence from the start. The pause makes the cut easy to find.
- Team: One Fold · College: Parul University (already filled in below).

Clip 1 · Intro (1:00)

Screen: your face, full screen. Smile before the first word.

Hi, I'm Priyanka Bharwani from Parul University, and I'm competing solo as team One Fold. Today I'll show you DilSe, which means "from the heart" in Hindi: my digital twin for heart patients.

Heart disease is one of the leading causes of death in India. Yet a heart patient usually sees a doctor only now and then: a clinic visit, an ECG, and then weeks or months of silence until the next visit.

In those weeks, many patients wear a smartwatch that records their heart rate, sleep and activity every single day. That data exists, but no doctor can watch thousands of patients' readings by hand.

So I built a twin that does the watching. It reads the patient's clinic ECG to decide how worried we should be. It learns each patient's own normal from their daily readings, starting on day one. And every morning it tells the doctor who needs attention today, why, and what to do next.

In the next few minutes I'll show you how it works, what is real and what is simulated, how well it did on patients it had never seen, and where its limits are.

Clip 2 · Problem and use case (1:45)

Screen: slide 1, the cover. Hold it for 3 seconds, then go to slide 2.

Between clinic visits, doctors lose sight of their heart patients, for three reasons. Care happens in occasional check-ups. Wearables record data every day, but nobody reads it. And a fixed rule like "resting heart rate above 100" doesn't work, because every person has their own normal: a rise of 8 beats can be serious for one patient and ordinary for another.

The challenge asks us to combine static clinical data with a continuous wearable stream, and to flag high-risk patients for the doctor.

Screen: slide 3.

My use case is remote follow-up of heart patients between clinic visits, in four steps.

One: at the clinic visit, my AI model reads the patient's ECG and labels them high risk or low risk. An ECG is one of the most common and affordable heart tests in Indian clinics, so the twin starts from data almost every clinic already has.

Two: every day, their resting heart rate, heart-rate variability, sleep and steps are compared with their own normal.

Three: the risk level tunes the alarm. High-risk patients get a more sensitive alarm, while the total number of false alarms stays within a fixed budget.

Four: every morning, the doctor opens a worklist sorted into Critical, Concern and Stable, with a reason and a suggested next step for each patient.

This tool supports triage. It gives no diagnosis and no treatment advice. The doctor always decides.

Clip 3 · Data, data flow and architecture (2:00)

Screen: slide 4.

The twin uses two streams of data, and I keep them clearly apart: one is real, one is simulated.

Stream one is real. It is PTB-XL, a public dataset on PhysioNet with 21,799 twelve-lead ECGs from 18,869 patients, each labelled by cardiologists. I used the official split by patient, so no patient appears in both training and testing.

Stream two is simulated, because no open dataset has both a clinic ECG and months of smartwatch data for the same people. I generated 60 days of readings for each patient, but every setting comes from real wearable data, PMData and MMASH, or from published studies. And the simulator only sees a patient's age and sex, never their ECG result.

Screen: slide 5.

This is the data flow diagram at level zero. Two sources feed one process, DilSe: the clinic ECG with age and sex, and the patient's daily wearable readings. There is one user, the doctor, with one flow in, selecting patients and adding reviews and notes, and one flow out, the worklist, alerts, next steps and a summary.

Screen: slide 6.

Inside, on the left, the real ECG goes into my AI model, four neural networks averaged, which gives a risk level. On the right, the daily readings are compared with the patient's own normal, which the twin learns from day zero. An alarm fires when readings stay unusual for several days in a row. In the middle the streams meet twice: the risk level sets how sensitive the alarm is, and risk plus alarm gives one daily status.

Clip 4 · The app at a glance (0:30)

Screen: slide 7.

This is DilSe, the app the doctor sees. On the left, the morning worklist with what changed and a suggested next step for every patient. On the right, a patient page with a one-line summary and the twin's profile of that patient. I'll show it live in a few minutes.

Clip 5 · ECG model and results (2:15)

Screen: slide 8.

The ECG model answers one question: is this ECG normal, or does it show an abnormality, such as a heart attack pattern, ST or T-wave changes, a conduction problem, or an enlarged heart muscle?

It is a one-dimensional convolutional neural network. It reads the raw signal from all twelve leads and learns the patterns itself, from small shapes like spikes and slopes up to whole

heartbeats. It outputs one number, the probability that the ECG is abnormal. It is small, about 190,000 parameters, and trains on an ordinary laptop.

I let experiments make the design choices, and I wrote down each decision rule before running it. 100 samples per second worked as well as 500 and trained five times faster. A bigger ResNet did not beat the simple network by the margin I set in advance. And adding age and sex made no measurable difference, so the score uses the ECG alone. The final score is the average of four networks.

Screen: slide 9.

On 2,145 test patients the model had never seen, real ECGs, it reaches an AUROC of 0.938. In plain words: pick one abnormal and one normal ECG at random, and about 94% of the time the model gives the abnormal one the higher score. Age and sex alone reach only 0.755.

At the cut-off I fixed in advance, it catches 80% of abnormal ECGs and correctly clears 91% of normal ones. So it misses about one in five, which is why the app warns that a low-risk result does not rule anything out.

Clip 6 · Wearable simulation and monitor (2:00)

Screen: slide 10.

Each simulated day follows a simple recipe: the person's usual level, plus a weekday effect, plus a day-to-day wobble with memory, so a high reading today makes a high one tomorrow more likely, just like in real people. Every number has a source: real Fitbit data from PMData, and published studies, including a figure of about 4,300 steps a day for India.

To test the monitor, half the patients get a deterioration episode: their resting heart rate rises by 3, 5, 8 or 12 beats per minute over a few days, while heart-rate variability, sleep and steps drop.

Screen: slide 11.

The monitor asks every day: is this patient unusual compared with their own normal, in the unhealthy direction, and is it lasting? It uses the patient's own four weeks of data, skipping the most recent week so a slow decline can't hide in its own baseline.

I tuned it on validation patients only, with a limit of half a false alarm per patient per month. It caught 93% of the 5-beat episodes, against 85% for a fixed baseline. A simple rule like "heart rate 10 beats above usual for 3 days" caught only 29% on average, against 90% for my monitor.

Clip 7 · The twin learns from day one (1:15)

Screen: slide 12.

There was one weakness. That monitor needs 35 days of data before it can judge anything, so a patient discharged today would be unmonitored for five weeks.

So the twin now learns from day one. On day zero it starts from population values for people like this patient, so its expected range is wide: about plus or minus 8 beats. Each day of the

patient's own data narrows it, to under 3 beats by day ten. From day 35 the original baseline takes over, unchanged.

I wrote the pass and fail rules before running it, and all three passed. Early episodes caught within two weeks rose from 11% to 87%, false alarms stayed within budget, and on real people the alarm rate was within the limit. One honest caveat: the synthetic gain is optimistic, because the starting values come from the same recipe as the simulator.

Clip 8 • The twin: combining the streams (1:00)

Screen: slide 13.

This is where the two streams become one personalised twin. The ECG risk level sets how closely we watch: a high-risk patient gets an alarm sooner. And this table gives the daily status. Low risk with no alarm is Stable. High risk or an alarm, but not both, is Concern. High risk with an alarm is Critical.

On the test patients, this moves attention towards high-risk patients at the same overall false-alarm rate. Because my simulated episodes don't depend on the ECG, this shows how attention is redistributed, not that it improves outcomes.

Clip 9 • Live demo of DilSe (4:15)

Screen: the app in your browser, Worklist open. Demo controls: Replay day 59, Show simulation answers off.

This is DilSe. Everything here is the 2,145 test patients the model never saw during development.

At the top, today's counts: 449 Critical, 24 of them new since yesterday; 980 Concern; and 716 Stable. Each row shows the patient's clinic ECG risk, a 14-day heart-rate trend, what changed in clinical units, and a suggested next step.

Screen: turn on "New today only", then turn it off.

The doctor can see just the patients who got worse since yesterday.

Screen: set Replay day to 40. Click "Patient" in the sidebar, choose PT-00160.

This is PT-00160, a 68-year-old woman, on day 40. The summary at the top tells me the situation: high-risk clinic ECG, 85%, and her readings are mostly within her usual range. So her status is Concern, and the suggested next step is to review the trend again in three days.

Screen: point at the three profile cards.

This is the twin's profile of her: her ECG risk and how far the four models agree, what the twin knows about her and her alert setting, and today's readings against her usual values.

Screen: drag Replay day slowly from 40 to 45.

Look what happens. Her resting heart rate starts to climb. On day 45 she moves to Critical, marked new today. The summary now says her readings have stayed away from her usual

range, resting heart rate up 4 beats, and the next step is to contact her within 24 hours. The status history at the bottom records the change.

Screen: turn on "Show simulation answers", then turn it off.

For this demo only, I can reveal the simulation's answer. Her episode started on day 41, so the twin caught it four days later, with no false alarm before it. A real doctor would never see this switch.

Screen: type a note, "Phoned patient, feels breathless on stairs", click "Mark as reviewed", then "Download summary (PDF)" and show the PDF for 3 seconds.

The doctor can add a note, mark her as reviewed, and download a one-page summary for her file or a referral.

Screen: open "12-lead clinic ECG" with the shading on.

This is her real twelve-lead ECG. The shading marks where small changes would move the risk score most. It is not a diagnosis, and I checked that it reflects what the model learned.

Screen: drag Replay day to 0, then to 10, then to 20. Point at the blue band and "What the twin knows".

Now the learning twin. On day zero the twin knows only population values: the blue band is wide and it is 0% learned. By day ten the band has narrowed around her own readings and it is 68% learned; by day twenty, 82%. She is monitored from the start, not from week five.

Screen: choose PT-00079. Set Replay day to 48, then 49.

A second patient. PT-00079 has a low-risk ECG. On day 48 he is Stable, but the summary already says it is watching his resting heart rate, up 10 beats. On day 49 the change has lasted, he moves to Concern, and the next step is to check in within three days. So the wearable stream alone can bring a low-risk patient to the doctor.

Screen: click "How it works" and scroll once.

Finally, How it works explains the status, the test results and the limits, in plain words.

Clip 10 · Integrity, limits and next steps (1:45)

Screen: slide 14.

How do you know these numbers are honest? I wrote down the decision rule for every comparison before running it: 28 logged decisions. One idea, an ECG heart-age model, failed its own rule, so I left it out of the app and reported it anyway. All tuning used validation patients, and the test set was opened once, by a script that locks itself. Thirteen automated tests check for leakage, look-ahead and unchanged settings. And I disclose one slip: during a dry run I saw the simulated test results early. Nothing was changed afterwards.

Screen: slide 15.

The limits. The ECG model detects a current abnormality; it does not predict a future event. The ECGs come from a German hospital, so performance on Indian patients is unknown. It misses about one abnormal ECG in five. All wearable data is simulated, and the learning twin's synthetic gain is optimistic. This is a research prototype, not a medical device.

Screen: slide 16.

The next steps follow from those limits: validate the ECG model on Indian patients' ECGs with a partner hospital, replace the simulated stream with real paired data, link to outcomes so it can predict future events, and add lab results and diagnosis history.

Clip 11 · Close (0:45)

Screen: your face, full screen.

To sum up: a real ECG model, tested once on patients it had never seen, sets each patient's risk. A twin that learns each patient's normal from day one watches their daily readings. And DilSe turns both into one status, a reason and a next step a doctor can act on, every morning.

Everything is in my GitHub repository, with a live demo link: the code, the decision log, the results and the tests, open source under the MIT licence.

Thank you for watching. I'm Priyanka Bharwani, team One Fold, from Parul University.

Checklist before recording

- ☐ Team One Fold and Parul University filled in, in this script, on the deck's cover and in the README (done)
- ☐ Phone or webcam held sideways (landscape), at eye level, facing a window or lamp
- ☐ Earphones with a mic, quiet room, 10-second sound test listened back
- ☐ Notifications off, laptop plugged in, other tabs closed, browser zoom 110–125%
- ☐ App running (streamlit run dashboard.py) or the live link open; Replay day 59, simulation answers off
- ☐ Demo path rehearsed once: New today only · PT-00160 day 40 → 45 · answers on/off · note, review, PDF · ECG · day 0 → 10 → 20 · PT-00079 day 48 → 49 · How it works
- ☐ Each clip recorded as its own file; joined video checked at 15–20 minutes
- ☐ Uploaded to YouTube as **Unlisted**, link opened in a private window, link added to the README and slide 16